

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights generally operate using fixed timers, while intelligent systems can use information from the road to adapt signal timing. This paper presents a local-AI-based traffic light architecture in which street cameras provide traffic information to smart pole computers for local processing. The system can also forward selected information to a city cloud server for storage and wider analysis. The proposed approach also considers emergency vehicle detection as an important research gap in existing traffic systems.

Index Terms—Smart Traffic Lights, Local AI, Traffic Signal Control, Machine Vision, Emergency Vehicle Detection

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded.

Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Instead of changing lights only according to a clock, the system uses information from the road. Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions.

Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

II. LITERATURE REVIEW

Previous research has investigated AI-based traffic signal control and adaptive traffic-light systems.

Rasheed et al. reviewed deep reinforcement learning approaches for traffic signal control and discussed the growing application of AI techniques in traffic management [1].

Gorodokin et al. investigated the optimization of adaptive traffic light control modes based on machine vision, demonstrating the importance of camera-based traffic information in adaptive signal control [2].

The literature therefore indicates that traffic signals can use real-time road information instead of relying only on fixed timers.

III. SYSTEM COMPARISON

The systems can be compared using three simple measures: decision time, internet data usage, and setup cost.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision	Internet	Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

IV. SYSTEM ARCHITECTURE

The proposed architecture follows four steps. First, street cameras capture traffic conditions. Second, the video information is sent to smart pole computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the city cloud server for storage and wider analysis. Fourth, the decision is sent to the traffic lights, which change their signals.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

V. SYSTEM FLOWCHART

The flowchart of the proposed architecture is shown in Fig. 1.

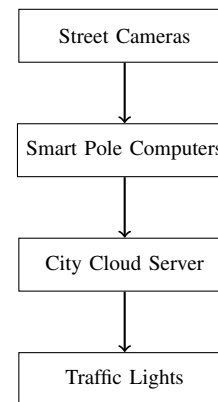


Fig. 1. Flowchart of the Proposed Traffic-Light System

VI. RESEARCH GAP

Existing traffic systems can work well for regular daily traffic, especially when traffic patterns are predictable. However, an important emergency problem remains: many systems do not have a built-in way to automatically clear a path for a speeding ambulance.

This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby signals to create a clear route.

VII. MATHEMATICAL MODEL

The traffic density can be represented using the number of vehicles detected by the camera:

$$D = \frac{N_v}{L} \quad (1)$$

where N_v represents the number of detected vehicles and L represents the road length considered.

The system can use the detected traffic information to determine the signal decision.

For example:

$$T = T_{\min} + kD \quad (2)$$

where T is the calculated signal time, $T_{\min}$ is the minimum signal time, D is the traffic density, and k is a system constant.

VIII. ARRAY REPRESENTATION

The traffic information detected from different road lanes can be stored in an array:

$$A = [a_1, a_2, a_3, \dots, a_n] \quad (3)$$

where each element represents the number of vehicles detected in a particular lane.

For multiple lanes, the information can be represented as a matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad (4)$$

IX. PROPOSED ALGORITHM

The basic operation of the proposed system can be represented as follows:

- 1) Capture traffic information using street cameras.
- 2) Send the information to the smart pole computer.
- 3) Process the traffic information locally.
- 4) Estimate the traffic condition.
- 5) Determine the appropriate traffic-light decision.
- 6) Send the decision to the traffic lights.
- 7) Store selected information in the city cloud server.

X. RESULTS AND DISCUSSION

The three systems can be compared based on their decision time, internet usage, and setup cost as shown in Table I.

The local smart-camera approach uses local processing and therefore can reduce the amount of information sent over the internet. This architecture may also allow faster decisions by processing information closer to the road.

The use of machine vision for adaptive traffic-light control is supported by previous research [2], while broader research has examined AI-based traffic signal control [1].

XI. CONCLUSION

Smart traffic-light systems can use real-time traffic information to improve the way traffic signals respond to changing road conditions. The proposed local-AI architecture uses street cameras, smart pole computers, a city cloud server, and traffic lights.

Local processing can reduce dependence on continuous internet communication and may allow quicker traffic decisions. An important research gap is the automatic detection of emergency vehicles and coordination of traffic signals to create a clear route.

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